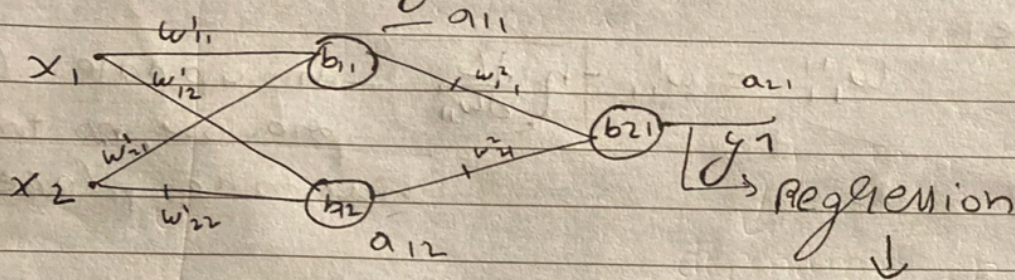


Weight Initialization

What not to do?

Case 1 $\rightarrow$ Zero Initialization



Activation

- $\rightarrow$ relu
- $\rightarrow$ Sigmoid
- $\rightarrow$ tanh

1] Relu Activation function

$$\text{Relu} = \max(0, z)$$

$$\text{let } w=0 \text{ and } b=0$$

$$\text{So } z_{11} = w'_{11}x_1 + w'_{21}x_2 + b_{11} = 0$$

$$\text{So } a_{11} = \max(0, z_{11}) = 0$$

$$z_{12} = w'_{12}x_1 + w'_{22}x_2 + b_{12} = 0$$

$$a_{12} = \max(0, z_{12}) = 0$$

$$a_{11} = a_{12} = 0$$

$\downarrow$ So there derivatives will also be 0

$$w'_{11} = w_{11} = \left[\eta \frac{dL}{dw_{11}} \right] 0$$

$$w'_{11} = w_{11}$$

$$\text{no update } \frac{dL}{dw_{11}} = \frac{dL}{dy^1} \times \frac{dy^1}{da_{11}} \times \frac{da_{11}}{dz_{11}} \times \frac{dz_{11}}{dw_{11}}$$

2) tanh

$$a_{11} = \frac{e^{z_{11}} - e^{-z_{11}}}{e^{z_{11}} + e^{-z_{11}}} \Rightarrow \frac{e^0 - e^0}{e^0 + e^0} = \frac{0}{1} = \underline{\underline{0}}$$

$$z_{11} = w_{11} n_1 + w_{21} n_2 + b = 0$$

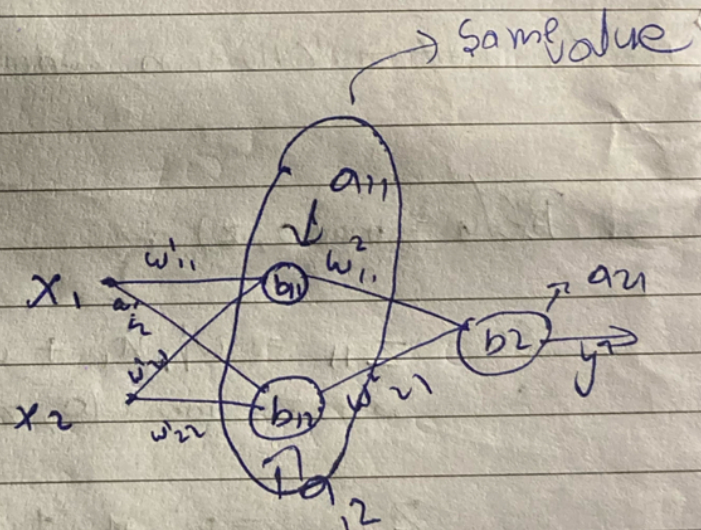
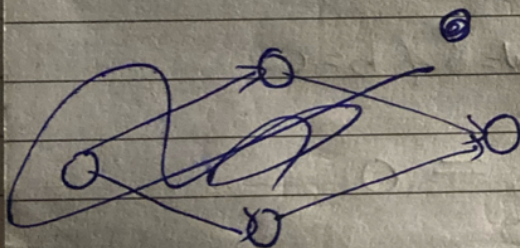
Same

$$w_{11}'' = w_{11} - \eta \left[\frac{dL}{dw_{11}} \right] \rightarrow \text{no update like previous, no training will take place}$$

Sigmoid

↳

$$\frac{1}{1 + e^{-x}}$$



$$z_{11} = w_{11} n_1 + w_{21} n_2 + b_{11} = 0 \quad [w \text{ and } b = 0]$$

$$\boxed{z_{11} = 0}$$

why activation function is $a_{11} = \frac{1}{1 + e^{-z}} = \frac{1}{1 + 1} = 0.5$

$$z_{12} = w_{12} n_1 + w_{22} n_2 + b_{12} = 0 \quad [w \text{ and } b = 0]$$

$$a_{12} = \frac{1}{1 + e^{-z}} = \frac{1}{2} = 0.5$$

So, After observation $a_{11} = a_{12} = 0.5$

If there are any number of neuron in that layer, they will still have some value, So my Neural Network will start behaving like perceptron because all layer's ~~all~~ neuron in layer 1 will share same value

Weights & biases of every neuron will be same.

$$\checkmark \frac{\partial L}{\partial w'_{11}} = \frac{\partial L}{\partial y^1} \frac{\partial y^1}{\partial a_{11}} \times \frac{\partial a_{11}}{\partial z_{11}} x_1$$

$$\checkmark \frac{\partial L}{\partial w'_{12}} = \frac{\partial L}{\partial y^1} \frac{\partial y^1}{\partial a_{12}} \times \frac{\partial a_{12}}{\partial z_{12}} x_1$$

→ Same output
because $a_{11} = a_{12}$
 $z_{11} = z_{12}$
So some new weights

$$\checkmark \frac{\partial L}{\partial w'_{21}} = \frac{\partial L}{\partial y^1} \frac{\partial y^1}{\partial a_{11}} \frac{\partial a_{11}}{\partial z_{21}} x_2$$

$$\checkmark \frac{\partial L}{\partial w'_{22}} = \frac{\partial L}{\partial y^1} \frac{\partial y^1}{\partial a_{12}} \frac{\partial a_{12}}{\partial z_{22}} x_2$$

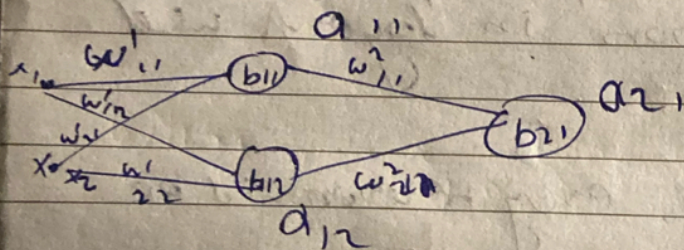
Same output
So, new same same
new weights

So never initialize weights & biases with zero for any activation function

Case 2 $\Rightarrow$ $w_{0n} = 0$ constant value

$$w = 0.5, b = 0.5$$

let the Activation function be Sigmoid



$$Z_{11} = w_{11}x_1 + w_{21}x_2 + b_{11} \neq 0$$

So

$$a_{11} = \max(0, Z_{11}) \neq 0$$

if w & b are
+ve

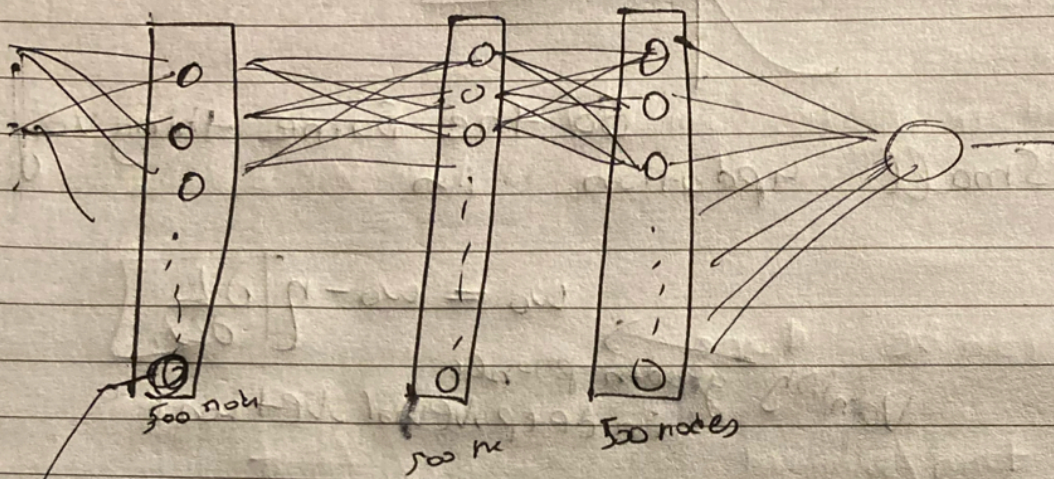
$$Z_{12} = w_{12}x_1 + w_{22}x_2 + b_{12} \neq 0$$

$$a_{12} = \max(0, Z_{12}) \neq 0$$

$a_{11} = a_{12}$ because weights & biases are same
& input also same so

So weight & biases of from one input
will be same like Sigmoid for zero initialization
This will same for other, Sigmoid & tanh.

Case 3 $\Rightarrow$ Random weight Initialization (Small)



$x_1, \dots, x_{500} | y$

1000×500

`np.random.randn(500, 500) * 0.01`

Small

$bias = 0$

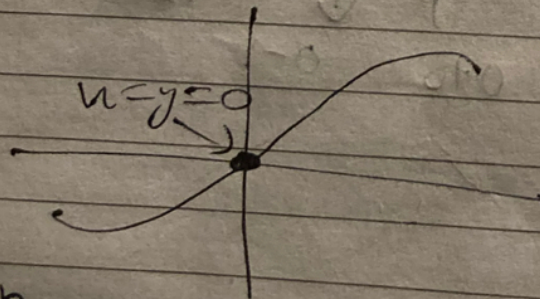
$$z = \sum w_i u_i$$

$$x_i \rightarrow -1 + 0.1$$

$$\rightarrow 0.0007$$

z is small number

tanh



$\rightarrow$ all weights will be

very close to zero

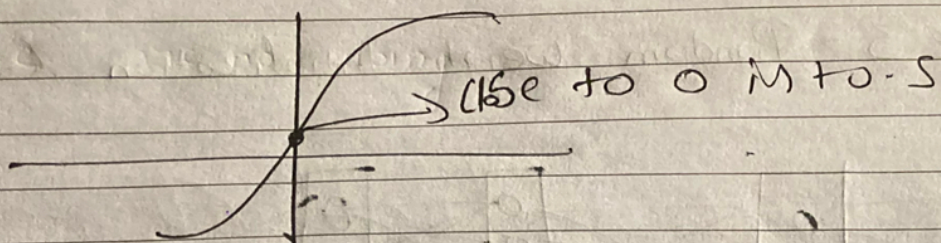
vanish high gradient

gradient

$$\rightarrow \text{so } w_0 = w_0 - \eta \frac{\partial L}{\partial w_0}$$

$\rightarrow$ No updation

Sigmoid



So the same thing very small updation in

$$w_0 = w_0 - \eta \left[\frac{\partial L}{\partial w_{i,j}} \right]$$

Vanishing gradient problem in deep Neural Network

ReLU

It will not reflect Vanishing gradient but need many epochs near 2000

case 4) Random Initialization with large values

$$\text{np.random.randn}(500, 500) \rightarrow [0 \rightarrow 1]$$

$$\sum w_i x_i \rightarrow 500 \text{ nodes} \rightarrow 100 - 500$$

range of weight which is very big

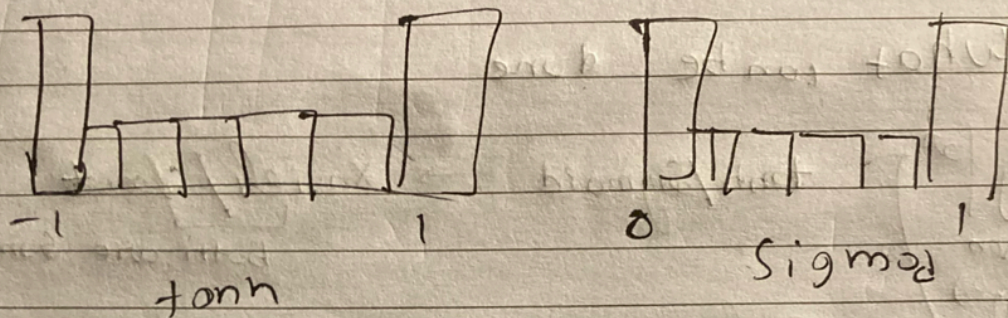
0 to 0-1

tanh / Sigmoid

$$\sum w_i x_i \rightarrow 100$$

↳ Saturate

It will go for high values of it -1 on 1 in tanh on 0-1 in Sigmoid



Relu

↓ No vanishing gradient but

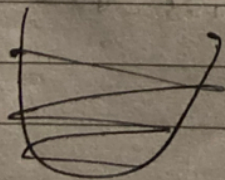
if

$$\sum w_i x_i \rightarrow 250$$

Output

then activation function will also be 250

So when we calculate GD the gradient will be more



→ not stable answers we will get

What we are not doing

- 1) No zero
- 2) No constant
- 3) Random values (small) $\times 0.01 \rightarrow$ VGP, Slow convergence
- 4) Random values (Large) $\times 1 \rightarrow$ VGP, Exploding GP, High weights

What can be done

only for hidden layers $\left\{ \begin{array}{l} \text{tanh/Sigmoid} \rightarrow \text{Xavier/Glorot} \\ \text{Relu} \rightarrow \text{he} \end{array} \right.$
both are same

Glorot $\rightarrow$ np.random.randn(10, 10) * $\left(\frac{1}{n} \right)$
variance
input on that node of hidden layer
 $\frac{1}{\sqrt{\text{fan in}}}$
if $n_T \downarrow w_T$
 $n_H \downarrow w_H$

he $\rightarrow$ np.random.randn(10, 10) * $\left(\frac{2}{n} \right)$

These are Normal distribution

Uniform distribution

Xavier uniform $\sim [-limit, limit]$

$$limit = \sqrt{\frac{6}{f_{on in} + f_{on out}}}$$

he uniform

$[-limit, limit]$

$$limit = \sqrt{\frac{6}{f_{on in}}}$$

By default in keras is
Heimdahl $\rightarrow$ Xavier uniform